

MISC02 — Continuous Talent Intelligence & Skill Growth

Proposed Advanced Features

1. AI-Driven Evidence Extraction & RAG Pipeline

Instead of relying on manual data entry, build an automated ingestion engine.

- **Automated Skill Tagging:** Integrate with GitHub, Jira, or internal project management tools to automatically parse project results.
- **Evidence Justification Engine:** Utilize LangChain and ChromaDB to build a Retrieval-Augmented Generation (RAG) pipeline. When the system suggests a next action, the LLM can pull specific code commits, test questions missed, or project roadblocks from the vector database to write a highly contextual justification.

2. Predictive Analytics & Skill Decay Modeling

Move beyond historical tracking into predictive insights.

- **Skill Half-Life Forecasting:** Implement machine learning models (e.g., XGBoost) to calculate the decay rate of a skill. If a user hasn't committed code in a specific language or completed a related course in six months, the system proactively warns of potential skill degradation before a drop in performance occurs.
- **Dynamic Confidence Scoring:** Calculate the confidence score dynamically based on data recency, the weight of the assessment, and data volume.

3. Next-Action Recommendation Engine

Enhance the suggested next action feature with hyper-personalized learning paths.

- **Micro-Learning Routing:** Connect the platform to external APIs. If a skill is labeled as getting worse, the system directly links to the exact module or documentation page needed to bridge the gap.
- **Internal Mentorship Matching:** If a manager identifies a team member needing support, the system can automatically query the database to find another employee with an improving or high-proficiency label in that exact skill to suggest a peer-mentorship pairing.

4. UI/UX and Dashboard Enhancements

- **Interactive Skill Graphs:** Build dynamic, interactive timeline views using React.js or Flutter, allowing users to hover over data points to see the exact test score or project result that influenced the trend line.
- **Role-Specific Action Centers:** For the Learner/Employee role, emphasize personal growth streaks and gamification. For the Manager/Mentor role, focus on high-level aggregation, such as a Team Skill Matrix heatmap.

Development Roadmap

Phase 1: Foundation & Architecture (Weeks 1-2)

- Design database schema (Users, Skills, Evidence Sources, Logs).
- Develop secure authentication for Learner and Manager roles.
- Build CRUD REST APIs (FastAPI) for manual entry of test scores, projects, and courses.

Phase 2: Core Logic & Analytics (Weeks 3-4)

- Implement the algorithm to classify skills as improving, flat, or declining.
- Develop the mathematical logic for the confidence score based on data points.
- Build the initial Learner dashboard UI to visualize the timeline view.

Phase 3: AI & Evidence Integration (Weeks 5-6)

- Integrate local LLMs (via Ollama) or cloud APIs.
- Set up ChromaDB to store and retrieve unstructured project/course data.
- Build the prompt templates to generate evidence-backed next actions.

Phase 4: Manager View & Polish (Weeks 7-8)

- Develop the Manager dashboard for reviewing team skill trends.
- Implement role-based access control (RBAC) to ensure managers only see their respective teams.
- Refine UI/UX animations and state management.

Phase 5: Deployment & Integrations (Weeks 9-10)

- Deploy backend to Render or Google Cloud and frontend to Vercel/Firebase.
- Build webhooks for external integrations (e.g., automatically pulling scores from an external testing platform).
- Final user acceptance testing.